

# NATIONAL OCCUPATIONAL STANDARD FOR FOOD SCIENTIST



NOS.FS.01  
FIRST EDITION

## APPROVING AUTHORITY

This National Occupational Standard (NOS) has been prepared and published under the authority of the Zambia Qualifications Authority Board on 15<sup>th</sup> February, 2024.

## ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority (ZAQA) Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide for the registration and accreditation of qualifications; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through the various sector specific National Occupational Standards Development Teams (NOSDTs).

## REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards (NOS) shall be revised every 5 years, or whenever it is deemed necessary, by the issuance of either amendments or of revised editions. It is important that users of the NOS ascertain that they are in possession of the latest amendments or editions.

## NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Manufacturing NOSDT, upon which the following organisations were represented:

1. Biomedical Society of Zambia (BMSZ)
2. Copperbelt University (CBU)
3. Engineering Institution of Zambia (EIZ)
4. Good Time Steel Company Zambia Limited (GTS)
5. Lusaka Vocational and Technical College (LVTC)
6. Makeni Ecumenical Centre (MEC)
7. Ministry of Commerce, Trade and Industry (MCTI)
8. Ministry of Health (MoH)
9. National Institute for Scientific and Industrial Research (NISIR)
10. Northern Technical College (NORTEC)
11. University of Lusaka (UNILUS)
12. University of Zambia (UNZA)
13. Zambia Association of Manufacturers (ZAM)
14. Zambia Forestry College (ZFC)
15. Zambia Qualifications Authority (ZAQA) – Secretariat

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## TABLE OF CONTENTS

|                                                                     |    |
|---------------------------------------------------------------------|----|
| FOREWORD.....                                                       | iv |
| JUSTIFICATION .....                                                 | iv |
| ACRONYMS AND ABBREVIATIONS .....                                    | v  |
| GLOSSARY OF TERMS .....                                             | vi |
| 1. OVERVIEW.....                                                    | 1  |
| 2. SCOPE.....                                                       | 3  |
| 3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES).....          | 3  |
| 4. UNITS AND ELEMENTS .....                                         | 3  |
| 5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS .....                  | 22 |
| 6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES.....              | 22 |
| 7. WORKING CONDITIONS/ENVIRONMENT .....                             | 22 |
| 8. PARTIES INVOLVED/INTERACTING WITH THE JOBHOLDER OR TRAINEE ..... | 22 |
| 9. PHYSICAL DEMANDS ON THE BODY .....                               | 23 |
| ANNEX A .....                                                       | 24 |
| Criteria for Assessments based on this NOS.....                     | 24 |
| ANNEX B .....                                                       | 25 |
| NOS Version Control.....                                            | 25 |

## FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“develop and implement a national qualifications framework; register and accredit qualifications; and ensure that standards and registered qualifications are internationally comparable.”***

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through the various sector specific National Occupational Standards Development Teams (NOSDTs) with experts composed of representation from the appropriate authorities, government departments, consumer associations, regulators, industry, academia, and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Manufacturing National Occupational Standards Development Team in accordance with the laid down procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as the National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies, and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula, and learning programmes in various sectors where the occupation exists. In the Manufacturing sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

## JUSTIFICATION

A Food Scientist is critical in the promotion of food that is healthy, safe, and of good quality. His/her role is to assess the nutritional content of food, discover new food sources, and research new ways to make raw and processed foods safe and healthy for the market. This National Occupational Standard highlights the core knowledge, skills, competences, and personal attributes that Food Scientists must possess to be successful in their jobs.

## ACRONYMS AND ABBREVIATIONS

|           |                                                  |
|-----------|--------------------------------------------------|
| CS        | Core Skills                                      |
| FS        | Food Scientist                                   |
| FSSC 2200 | Food Safety System Certification                 |
| GMP       | Good Manufacturing Practices                     |
| HACCP     | Hazard Analysis Critical Points                  |
| ISO       | International Standards Organisation             |
| K         | Knowledge and Understanding                      |
| NOS       | National Occupational Standard                   |
| NOSDT     | National Occupational Standards Development Team |
| OK        | Organisational Context                           |
| OSHE      | Occupational Safety, Health and Environment      |
| PC        | Performance Criteria                             |
| PS        | Professional Skills                              |
| RK        | Regulatory Context                               |
| RPL       | Recognition of Prior Learning                    |
| S         | Skills                                           |
| TK        | Technical Knowledge                              |
| ZABS      | Zambia Bureau of Standards                       |
| ZAQA      | Zambia Qualifications Authority                  |
| ZQF       | Zambia Qualifications Framework                  |

## GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

**Core Skills/Generic Skills:** are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

**Disability:** is the physical or mental impairment that substantially limits one or more major life activities.

**Function:** is an activity necessary for achieving the key purpose of the sector, occupation or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of the NOS.

**Job Title:** defines a unique set of functions that together form a unique employment opportunity in an organisation.

**Knowledge and Understanding:** are statements which together specify the technical, generic, professional, and organisational specific knowledge that an individual needs in order to perform to the required standard.

**National Occupational Standards (NOS):** are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

**National Occupational Standards (NOS) Code:** is a unique reference code that identifies a NOS.

**National Occupational Standards Development Team (NOSDT):** means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

**Occupation:** is a set of job roles which perform similar/related set of functions in an industry.

**Organisational Context:** includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

**Performance Criteria:** are statements that together specify the standard of performance required when carrying out a task.

**Scope:** is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

**Sector:** is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy, whose components share similar characteristics and interests.

**Sub Sector:** is derived from a further breakdown based on the characteristics and interests of its components.

**Technical Knowledge:** is the specific knowledge needed to accomplish specific designated responsibilities.

**Unit Title:** gives a clear overall statement about what the incumbent should be able to do.

**Elements:** set out competences the incumbent should possess to carry out the day-to-day activities.



## 1. OVERVIEW

This is an introductory section providing a summary and specific information or commentary about the content of the NOS, and the targeted sector and occupation to help the user judge whether it is relevant to them.

|                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>                                                      | NOS.FS.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Occupation</b>                                                    | Food Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Job Title</b>                                                     | Food Scientist or Food Technologist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Job Description</b>                                               | <p>A Food Scientist is responsible for assessing the food's nutritional content, testing food quality, discovering new food sources, researching new ways to make raw and processed foods safe and healthy for the market. He/she also manages, operates, and maintains laboratory equipment, trains staff in food production related aspects, and document profiling.</p> <p>A Food Scientist is also responsible for ensuring adherence to the laws and regulations governing food processing, labelling, quality, safety, storage, and distribution of food products.</p> |
| <b>Job Purpose</b>                                                   | To develop, produce, process, preserve, and promote food which is healthy, safe, good quality, adaptable, and innovative.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>ZQF Level</b>                                                     | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Sector</b>                                                        | Manufacturing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Sub sectors</b>                                                   | Food Processing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Other Economic Sector(s) in which the Occupation is Practiced</b> | Tourism and Education                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Other Similar Jobs that can be Performed in the Occupation</b>    | Research Manager, Tutor/Lecturer/Trainer of Trainers<br>Quality Control Manager, Production Manager, Consultant, Food Analyst, Laboratory Scientist.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Minimum Educational Job Entry Qualification(s)</b>                | Bachelor's degree in Biology, Chemistry, Food Science or its equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Practicing License Requirements (if any)</b>                      | Advantageous to belong to the Zambia Association of Manufacturers, Food Science Association or other professional association related to food production                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Training/RPL</b>                                                  | <ol style="list-style-type: none"> <li>1. Awareness of the Food Industry standards, and rules and regulations, and their application.</li> <li>2. Awareness about Food Industry Regulators such as Zambia Bureau of Standards (ZABS), Zambia Compulsory Agency (ZCSA), Ministry of Health,</li> </ol>                                                                                                                                                                                                                                                                        |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | <p>Zambia Metrology Agency, Ministry of Local Government through Councils.</p> <p>3. Awareness about Food Industry statutory regulations such as Food Safety Act, and the Public Health Act, and other Statutory Instruments that affect the Food Industry such as Environmental Management Act, Metrology Act, Standards Act, etc.</p> <p>4. Food quality enhancement methods.</p> |
| <b>Minimum Job Entry Age</b>        | 21 years                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Prior Experience (Suggested)</b> | Minimum of 1-Year Internship                                                                                                                                                                                                                                                                                                                                                        |
| <b>Performance Criteria</b>         | As described in the Units under Section 4                                                                                                                                                                                                                                                                                                                                           |

## **2. SCOPE**

This NOS highlights the core competences, skills, fundamental knowledge and understanding, and personal attributes that Food Scientists are required to possess to be successful in their jobs.

## **3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)**

This job requires an individual to possess ability to apply and interpret national and international food safety and quality standards, such as standards set by the ZABS, Food Safety regulations, Codex Alimentarius, FSSC 2200, ISO, GMP. The individual is expected to communicate clearly and effectively, and use excellent reasoning as well as plan and prioritise their work. He/she must maintain personal hygiene, have good moral integrity, exceptional technical and problem-solving skills. He/she must also be creative, courteous, self-motivated; a great team worker, quality consciousness, occupational health and safety orientated, physically fit, and able to attend to work for long hours if required.

## **4. UNITS AND ELEMENTS**

This National Occupational Standard is divided into 6 Units, representing the tasks that a jobholder should undertake in his/her day-to-day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

**UNIT 1** [This Unit covers the skills and knowledge required by a Food Scientist to provide leadership, direction, and execution of all plant food safety functions to support the company's food safety goals]

|                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit No.</b>                                                             | <b>01</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Unit Title</b>                                                           | <b>Leadership and Execute Food Safety Functions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description</b>                                                          | This Unit describes the skills and knowledge required by a Food Scientist to provide leadership, direction, and execution of all related food safety functions to support the company's product food safety goals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Scope</b>                                                                | <p>This Unit covers the following:</p> <ul style="list-style-type: none"> <li>• Planning and Supervising Food Safety Functions for Food Safety Goals.</li> <li>• Assessing Non-conformities and Follow-up on Corrective Action.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Performance Criteria (PC) with respect to the Scope</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Element</b>                                                              | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Planning and Supervising Food Safety Functions for Food Safety Goals</b> | <p>To be competent, the individual must be able to:</p> <p>PC1. Assemble and supervise a representative food safety team.</p> <p>PC2. Develop, implement, and review organisation food safety documents to identify gaps related to national and international standards.</p> <p>PC3. Plan for internal and external quality controls.</p> <p>PC4. Establish periodic internal and external audits.</p> <p>PC5. Initiate corrective action to the identified gaps.</p> <p>PC6. Setup a communication strategy for food safety related Information.</p>                                                                                                                                                                                                                                                                                                                                         |
| <b>Assessing Non-Conformities and Follow-up on Corrective Action</b>        | <p>To be competent, the individual must be able to:</p> <p>PC7. Measure and interpret food safety parameters.</p> <p>PC8. Analyse/assess the gaps in meeting the national and international standards.</p> <p>PC9. Evaluate key performance indicators in the food safety systems.</p> <p>PC10. Identify internal and external quality controls.</p> <p>PC11. Arrange internal and external audits.</p> <p>PC12. Identify potential pests and log in an inventory.</p> <p>PC13. Plan food handlers to ensure that all staff are in good health.</p> <p>PC14. Create a pest control programme, (i.e., include all elements that are involved in the implementation of full scale GMP food safety management system).</p> <p>PC15. Plan and organise appropriate PPEs for all staff.</p> <p>PC16. Design a cleaning programme for the facility.</p> <p>PC17. Design a product recall system.</p> |
| <b>Knowledge and Understanding (K)</b>                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>A. Organisational Context (Knowledge of the company/</b>                 | <p>The individual on the job must demonstrate knowledge and understanding of organisational:</p> <p>OK1. Food safety goals.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>organisation and its processes)</b>                            | OK2. Food quality goals.<br>OK3. Food safety policies, rules and regulations.<br>OK4. Standard operating procedures.<br>OK5. Food product requirements and specifications.                                                                                                                                                                                                                                                         |
| <b>B. Technical Knowledge</b>                                     | The individual on the job must demonstrate knowledge and understanding of food:<br><br>TK1. Quality control techniques.<br>TK2. Safety management systems.<br>TK3. Processing techniques and product development.<br>TK4. Packaging, storage, and distribution chain requirements.                                                                                                                                                 |
| <b>C. Regulatory Context (knowledge of rules and regulations)</b> | The individual on the job must demonstrate knowledge and understanding of the:<br><br>RK1. Public Health Act.<br>RK2. Food Safety Act and regulations.<br>RK3. Zambia Bureau of Standards Act and regulations.<br>RK4. Zambia Compulsory Standards Agency Act.<br>RK5. Environmental Management Act and regulations.<br>RK6. Occupational Health and Safety Act, and other related regulations.<br>RK7. Workers' Compensation Act. |
| <b>Skills (S)</b>                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>A. Core Skills/ Generic Skills</b>                             | <b>Writing Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                   | The individual on the job must be able to:<br><br>CS1. Write in English, and/or have the means to give comprehensible instructions on food-related issues.<br>CS2. Prepare and provide clear and simple instructions or details to co-workers.                                                                                                                                                                                     |
|                                                                   | <b>Reading Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                   | The individual on the job must be able to:<br><br>CS3. Read English, and/or be able to give instructions in the local language.<br>CS4. Read and interpret instrument analytical printouts or instructions provided for the required work.                                                                                                                                                                                         |
|                                                                   | <b>Oral Communication (Listening and Speaking Skills)</b>                                                                                                                                                                                                                                                                                                                                                                          |
| <b>B. Professional Skills</b>                                     | The individual on the job must be able to:<br><br>CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers.<br>CS6. Convey information clearly and concisely to co-workers and subordinates.                                                                                                                                                                                     |
|                                                                   | <b>Decision-Making</b>                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                   | The individual on the job must be able to:<br><br>PS1. Determine and report to the superior whether the work space and resources are adequate to produce a safe and good quality product.<br>PS2. Recommend to superiors on best practices of food safety                                                                                                                                                                          |

|  |                                                                                                     |
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|  | <b>Plan and Organise</b>                                                                            |
|  | The individual on the job should be able to:                                                        |
|  | PS3. Plan work and organise the required resources in coordination with team members and superiors. |
|  | PS4. Plan and organise quality control audits and inspections with specific regulatory agencies.    |
|  | PS5. Plan and organise efficient resources utilisation.                                             |
|  | <b>Customer Centricity</b>                                                                          |
|  | The individual on the job should be able to:                                                        |
|  | PS6. Manage relationships with customers with intent to satisfying their product requirements.      |
|  | <b>Problem-Solving</b>                                                                              |
|  | The individual on the job should be able to:                                                        |
|  | PS7. Resolve any conflicts within the team.                                                         |
|  | PS8. Coordinate with different departments to achieve smooth workflow.                              |
|  | PS 9. Share workload.                                                                               |
|  | PS10. Contribute to quality customer satisfaction.                                                  |
|  | PS11. Comply with regulators on food product quality and safety.                                    |
|  | PS12. Pay attention to detail.                                                                      |
|  | <b>Analytical Thinking</b>                                                                          |
|  | The individual on the job should be able to:                                                        |
|  | PS13. Analyse ideas and use logic to formulate appropriate action.                                  |
|  | PS14. Be flexible enough to apply new technologies.                                                 |
|  | <b>Critical Thinking</b>                                                                            |
|  | The individual on the job should be able to:                                                        |
|  | PS15. Identify and deal with emerging food safety non-compliances.                                  |

**UNIT 2** [This Unit covers the skills and knowledge required by a Food Scientist in assessing the nutritional content of food and researching new ways to make both raw and processed food nutritious]

|                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Unit No.</b>                                                                             | <b>02</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Unit Title</b>                                                                           | <b>Assess Nutritional Food Content and Research New Formulas to Make Nutritious Products</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Description</b>                                                                          | This Unit describes the skills and knowledge required by a Food Scientist to assess the nutritional content, safety of food, and researching new ways to make raw and processed food nutritious.                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Scope</b>                                                                                | <p>This unit covers the following:</p> <ul style="list-style-type: none"> <li>• Optimising Food Ingredient Formulation and Product Development.</li> <li>• Assessing the Nutritional Content and Determination of Contaminants.</li> <li>• Research Product Development.</li> </ul>                                                                                                                                                                                                                                                                                                                |
| <b>Performance Criteria (PC) with respect to the Scope</b>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Element</b>                                                                              | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Optimising Food Ingredient Formulation and Product Development</b>                       | <p>To be competent, the individual must be able to:</p> <p>PC1. Identify the critical ingredients of the products.</p> <p>PC2. Enhance product nutritional content within the confines of quality specifications and regulations.</p> <p>PC3. Coordinate with the marketing department to meet customer satisfaction in new products developed.</p>                                                                                                                                                                                                                                                |
| <b>Assessing Nutritional Content and Determination of Contaminants</b>                      | <p>To be competent, the individual must be able to:</p> <p>PC4. Measure and interpret nutritional parameters.</p> <p>PC5. Interpret product specifications in relation to setting local and international standards.</p> <p>PC6. Assess conformity of raw materials/ingredients and packaging quality to acceptable standards.</p> <p>PC7. Assess raw materials or ingredients for possible sources of contamination.</p> <p>PC8. Identify internal and external standard controls for the new product.</p> <p>PC9. Assess suitability and usability of testing kits relating to new products.</p> |
| <b>Research Product Development</b>                                                         | <p>To be competent, the individual must be able to:</p> <p>PC10. Develop new products.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Knowledge and Understanding (K)</b>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>A. Organisational Context (Knowledge of the company/ organisation and its processes)</b> | <p>The individual on the job must demonstrate knowledge and understanding of organisational food:</p> <p>OK1. Quality goals.</p> <p>OK2. Nutritional profile goals.</p> <p>OK3. Quality policies, rules, and regulations.</p> <p>OK4. Standard operating procedures.</p> <p>OK5. Product requirements and specifications.</p>                                                                                                                                                                                                                                                                      |

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| <b>B. Technical Knowledge</b>                                     | <p>The individual on the job must demonstrate knowledge and understanding of food:</p> <p>TK1. Quality control techniques.<br/>TK2. Nutritional profile techniques.<br/>TK3. Fortification techniques and product formulation.</p>                                                                                                                                          |
| <b>C. Regulatory Context (knowledge of rules and regulations)</b> | <p>The individual on the job must demonstrate knowledge and understanding of the:</p> <p>RK1. Food and Drugs regulations (e.g. SI 90 of 2001).<br/>RK2. Food Safety Act and other regulations.<br/>RK3. Zambia Bureau of Standards Act and other regulations.<br/>RK4. Zambia Compulsory Standards Agency Act.<br/>RK5. Public Health Act.<br/>RK6. Codex Alimentarius.</p> |
| <b>Skills (S)</b>                                                 |                                                                                                                                                                                                                                                                                                                                                                             |
| <b>A. Core Skills/ Generic Skills</b>                             | <b>Writing Skills</b>                                                                                                                                                                                                                                                                                                                                                       |
|                                                                   | <p>The individual on the job must be able to:</p> <p>CS1. Write in English, and/or be able to give simple instructions in the local language.<br/>CS2. Prepare and provide clear and simple instructions and details, to co-workers.</p>                                                                                                                                    |
|                                                                   | <b>Reading Skills</b>                                                                                                                                                                                                                                                                                                                                                       |
|                                                                   | <p>The individual on the job must be able to:</p> <p>CS3. Read English, and/or be able to give simple instructions in the local language used.<br/>CS4. Read and interpret sketches, drawings or instructions provided for the required work.</p>                                                                                                                           |
|                                                                   | <b>Oral Communication (Listening and Speaking Skills)</b>                                                                                                                                                                                                                                                                                                                   |
|                                                                   | <p>The individual on the job must be able to:</p> <p>CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers.<br/>CS6. Convey information clearly and concisely to co-workers.</p>                                                                                                                                       |
| <b>B. Professional Skills</b>                                     | <b>Decision-Making</b>                                                                                                                                                                                                                                                                                                                                                      |
|                                                                   | <p>The individual on the job must be able to:</p> <p>PS1. Determine and report to the superior whether the work space and resources are adequate to produce safe products.<br/>PS2. Recommend to superiors on best practices on food formulations to ensure safe and nutritious food supply.</p>                                                                            |
|                                                                   | <b>Plan and Organise</b>                                                                                                                                                                                                                                                                                                                                                    |
|                                                                   | <p>The individual on the job should be able to:</p> <p>PS3. Plan work and organise required resources in coordination with team members and superiors.<br/>PS4. Plan and organise quality control audits and inspections with specific regulatory agencies.<br/>PS5. Plan and schedule lab testing for compliance to regulatory standards.</p>                              |



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|  | PS6. Plan and organise efficient resources utilisation.                                            |
|  | <b>Customer Centricity</b>                                                                         |
|  | The individual on the job should be able to:                                                       |
|  | PS7. Manage relationships with the customers with intent to satisfying their product requirements. |
|  | <b>Problem-Solving</b>                                                                             |
|  | The individual on the job should be able to:                                                       |
|  | PS8. Resolve any conflicts within the team.                                                        |
|  | <b>Analytical Thinking</b>                                                                         |
|  | The individual on the job should be able to:                                                       |
|  | PS9. Analyse and convey to the superior and carry out remedial action.                             |
|  | <b>Critical Thinking</b>                                                                           |
|  | The individual on the job should be able to:                                                       |
|  | PS10. Identify and deal with or report violation of any safety norms which may lead to accidents.  |

**UNIT 3** [This Unit covers the skills and knowledge required by a Food Scientist to ensure that a specific food product meets existing food quality and safety specifications on the market]

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| <b>Unit No.</b>                                            | <b>03</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Unit Title</b>                                          | <b>Product Testing for Required Quality and Safety Specifications</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Description</b>                                         | This Unit describes the skills and knowledge required by a Food Scientist to ensure that specific food products meet the existing food quality and safety specifications on the market.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Scope</b>                                               | This unit covers the following; <ul style="list-style-type: none"> <li>• Food Sampling.</li> <li>• Food Testing and Analysis.</li> <li>• Food Packaging and Labelling.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Performance Criteria (PC) with respect to the Scope</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Element</b>                                             | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Food Sampling</b>                                       | To be competent, the individual must be able to: <p>PC1. Recognise the hazard critical control points of the specific product in the value chain from farm to fork.</p> <p>PC2. Identify the possible hazards and spoilage organisms.</p> <p>PC3. Categorise hazards into chemical, biological or toxicological.</p> <p>PC4. Determine appropriate sampling points and size.</p> <p>PC5. Determine sample size as it relates to existing food sampling protocols.</p> <p>PC6. Design a sampling plan.</p> <p>PC7. Employ appropriate sampling kits for internal and external quality control testing.</p> <p>PC8. Comply with appropriate transportation guidelines for external testing of products or raw materials.</p> |
| <b>Food Testing and Analysis</b>                           | To be competent, the individual must be able to: <p>PC9. Determine appropriate testing methods.</p> <p>PC10. Analyse food products with appropriate controls.</p> <p>PC11. Explain and interpret analytical results.</p> <p>PC12. Determine appropriate corrective measures when unsatisfactory results occur.</p> <p>PC13. Prepare an analytical report of the findings.</p> <p>PC14. Identify which proficiency testing schemes (external quality assessment schemes) to utilise to maintain specific product safety and quality processes.</p>                                                                                                                                                                          |
| <b>Food Packaging and Labelling</b>                        | To be competent, the individual must be able to; <p>PC15. Identify the ideal (food grade) packaging for specific products.</p> <p>PC16. Determine the shelf life of the products.</p> <p>PC17. Apply 'Best Before' and 'Sell By' dates appropriately.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Knowledge and Understanding (K)</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>A. Organisational Context (Knowledge of</b>             | The individual on the job must demonstrate knowledge and understanding of organisational food:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| <b>the company/<br/>organisation and<br/>its processes)</b>                       | OK1. Safety goals.<br>OK2. Quality goals.<br>OK3. Safety policies, rules and regulations.<br>OK4. Standard operating procedures.<br>OK5. Product requirements and specifications                                                                                                                                                                                         |
| <b>B. Technical<br/>Knowledge</b>                                                 | The individual on the job must demonstrate knowledge and understanding of;<br><br>TK1. Food sampling techniques for biological or chemical analysis.<br>TK2. Biological and chemical food testing techniques.<br>TK3. Food processing techniques.                                                                                                                        |
| <b>C. Regulatory<br/>Context<br/>(knowledge of<br/>rules and<br/>regulations)</b> | The individual on the job must demonstrate knowledge and understanding of the:<br><br>RK1. Food and Drugs regulations (SI 90 of 2001).<br>RK2. Food Safety Act and other regulations.<br>RK3. Zambia Standards Act and other regulations.<br>RK4. Zambia Compulsory Standards Agency Act.<br>RK5. Local Government Act and other regulations.<br>RK6. Codex Alimentarius |
| <b>Skills (S)</b>                                                                 |                                                                                                                                                                                                                                                                                                                                                                          |
| <b>A. Core Skills/<br/>Generic Skills</b>                                         | <b>Writing Skills</b>                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                   | The individual on the job must be able to:<br><br>CS1. Write in English, and/or be able to give simple instructions in the local language.<br>CS2. Prepare and provide clear and simple instructions details to co-workers.<br>CS3. Construct an analytical report or certificate of analysis.                                                                           |
|                                                                                   | <b>Reading Skills</b>                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                   | The individual on the job must be able to:<br><br>CS4. Read English, and be able to give instructions to subordinates.<br>CS5. Read, interpret results, and convey information to superiors.                                                                                                                                                                             |
|                                                                                   | <b>Oral Communication (Listening and Speaking Skills)</b>                                                                                                                                                                                                                                                                                                                |
| <b>B. Professional<br/>Skills</b>                                                 | The individual on the job must be able to:<br><br>CS6. Communicate effectively with superior and other co-workers on laboratory findings.<br>CS7. Convey information clearly and concisely to co-workers and customers where applicable.                                                                                                                                 |
|                                                                                   | <b>Decision-Making</b>                                                                                                                                                                                                                                                                                                                                                   |
| <b>B. Professional<br/>Skills</b>                                                 | The individual on the job must be able to:<br><br>PS1. Determine analytical parameters to sample and test.<br>PS2. Recommend to superiors on best practices in food sampling, analysis, and packaging.                                                                                                                                                                   |

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|  | <b>Plan and Organise</b>                                                                                                                 |
|  | The individual on the job should be able to plan and organise:                                                                           |
|  | PS3. Sampling plan, including required resources.                                                                                        |
|  | PS4. A work schedule coordinating the tasks for sampling and testing for compliance to regulations.                                      |
|  | <b>Customer Centricity</b>                                                                                                               |
|  | The individual on the job should be able to:                                                                                             |
|  | PS5. Coordinate with other internal units that relate directly with customer care, with intent to satisfying their product requirements. |
|  | <b>Problem-Solving</b>                                                                                                                   |
|  | The individual on the job should be able to resolve any:                                                                                 |
|  | PS6. Conflicts within the team.                                                                                                          |
|  | PS7. Conflicts with clients in relation to analytical reports.                                                                           |
|  | <b>Analytical Thinking</b>                                                                                                               |
|  | The individual on the job should be able to:                                                                                             |
|  | PS8. Analyse and convey to the superior any unsatisfactory results and non-conformities.                                                 |
|  | PS9. Discuss with superior any occurrence that may require change in type of sampling or analytical methods.                             |
|  | <b>Critical Thinking</b>                                                                                                                 |
|  | The individual on the job should be able to:                                                                                             |
|  | PS10. Identify and deal with or report violation of guidelines or specifications that may require product recall as a remedial measure.  |

**UNIT 4** [This Unit covers the skills and knowledge required by a Food Scientist to profile documentation for certification to national and international bodies to meet food quality and safety standards.

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| <b>Unit No.</b>                                                                             | <b>04</b>                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Unit Title</b>                                                                           | <b>National and International Bodies Certification</b>                                                                                                                                                                                                                                                                                                                                                                |
| <b>Description</b>                                                                          | This Unit describes the skills and knowledge required by a Food Scientist to ensure that specific food products are certified to meet the food quality and safety standards of national and international bodies.                                                                                                                                                                                                     |
| <b>Scope</b>                                                                                | This Unit covers the following: <ul style="list-style-type: none"> <li>• Identification of Applicable Standards with Certification Bodies.</li> <li>• Document Profiling and Document control.</li> </ul>                                                                                                                                                                                                             |
| <b>Performance Criteria (PC) with respect to the Scope</b>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Element</b>                                                                              | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Identification of Applicable Standards with Certification Bodies</b>                     | To be competent, the individual must be able to: <p>PC1. Recognise the applicable food safety and quality standards with local and international certification bodies.</p> <p>PC2. Identify standards and protocols required by the local and international certification agencies.</p> <p>PC3. Coordinate with certification bodies on requirements for inspections and audits.</p>                                  |
| <b>Document Profiling and Document Control</b>                                              | To be competent, the individual must be able to: <p>PC4. Develop data filling and storage methods.</p> <p>PC5. Develop specific screening and data management systems.</p> <p>PC6. Establish document review and control procedures.</p> <p>PC7. Establish and supervise team of staff to record and profile documents.</p> <p>PC8. Establish an archiving or culling/disposal system of old documents procedure.</p> |
| <b>Knowledge and Understanding (K)</b>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>A. Organisational Context (Knowledge of the company/ organisation and its processes)</b> | The individual on the job must demonstrate knowledge and understanding of the organisational food: <p>OK1. Safety goals.</p> <p>OK2. Quality goals.</p> <p>OK3. Safety policies, rules, and regulations.</p> <p>OK4. Standard operating procedures.</p> <p>OK5. Product requirements and specifications.</p>                                                                                                          |
| <b>B. Technical Knowledge</b>                                                               | The individual on the job must demonstrate knowledge and understanding of: <p>TK1. Record keeping techniques.</p> <p>TK2. Local and international certification bodies.</p>                                                                                                                                                                                                                                           |
| <b>C. Regulatory Context (knowledge of</b>                                                  | The individual on the job must demonstrate knowledge and understanding of the: <p>RK1. Food and Drugs regulations (e.g. SI 90 of 2001).</p> <p>RK2. Food Safety Act and other regulations.</p>                                                                                                                                                                                                                        |

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| <b>rules and regulations)</b>             | <p>RK3. Zambia Standards Act.</p> <p>RK4. Zambia Compulsory Standards Agency Act.</p> <p>RK5. Local Government Act and other regulations.</p> <p>RK6. Codex Alimentarius.</p> <p>RK7. ISO Standards.</p>                                                                                                                                 |
| <b>Skills (S)</b>                         |                                                                                                                                                                                                                                                                                                                                          |
| <b>A. Core Skills/<br/>Generic Skills</b> | <b>Writing Skills</b>                                                                                                                                                                                                                                                                                                                    |
|                                           | <p>The individual on the job must be able to:</p> <p>CS1. Write in English (at least working level) and/or be able to give simple instructions in the local language.</p> <p>CS2. Prepare and provide clear and simple instructions or details to co-workers.</p> <p>CS3. Construct an analytical report or certificate of analysis.</p> |
|                                           | <b>Reading Skills</b>                                                                                                                                                                                                                                                                                                                    |
|                                           | <p>The individual on the job must be able to:</p> <p>CS4. Read English, and be able to give instructions to subordinates.</p> <p>CS5. Read, interpret results, and convey information to superiors.</p>                                                                                                                                  |
|                                           | <b>Oral Communication (Listening and Speaking Skills)</b>                                                                                                                                                                                                                                                                                |
|                                           | <p>The individual on the job must be able to:</p> <p>CS6. Communicate effectively with superiors and other co-workers on data management.</p> <p>CS7. Convey information clearly and concisely to co-workers and customers where applicable.</p>                                                                                         |
| <b>B. Professional Skills</b>             | <b>Decision-Making</b>                                                                                                                                                                                                                                                                                                                   |
|                                           | <p>The individual on the job must be able to:</p> <p>PS1. Determine which type of data should be stored.</p> <p>PS2. Recommend to superiors on best practices of recording and storing data.</p>                                                                                                                                         |
|                                           | <b>Plan and Organise</b>                                                                                                                                                                                                                                                                                                                 |
|                                           | <p>The individual on the job should be able to:</p> <p>PS3. Plan and organise data management, including required resources.</p> <p>PS4. Plan and organise a work schedule coordinating the tasks for filing protocols.</p>                                                                                                              |
|                                           | <b>Customer Centricity</b>                                                                                                                                                                                                                                                                                                               |
|                                           | <p>The individual on the job should be able to:</p> <p>PS5. Coordinate with other internal units that relate directly with customer care with intent to satisfying their product requirements.</p>                                                                                                                                       |
|                                           | <b>Problem-Solving</b>                                                                                                                                                                                                                                                                                                                   |
|                                           | <p>The individual on the job should be able to:</p> <p>PS6. Resolve any conflicts within the team.</p> <p>PS7. Resolve any conflicts with clients in relation to analytical reports.</p>                                                                                                                                                 |

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|  | <b>Analytical Thinking</b>                                                                           |
|  | The individual on the job should be able to:                                                         |
|  | PS8. Analyse and convey to the superior any unsatisfactory data management issues.                   |
|  | PS9. Discuss with superior any occurrence that may require change in type of data profiling methods. |
|  | <b>Critical Thinking</b>                                                                             |
|  | The individual on the job should be able to:                                                         |
|  | PS10. Identify and deal with or report violation of guidelines or specifications on data management. |

**UNIT 5** [This Unit covers the skills and knowledge required by a Food Scientist to supervise and train staff on food quality and safety aspects]

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| <b>Unit No.</b>                                                                             | <b>05</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Unit Title</b>                                                                           | <b>Food Quality and Safety Staff Supervision and Training</b>                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Description</b>                                                                          | This Unit describes the skills and knowledge required by a Food Scientist to supervise and train staff on food quality and safety aspects.                                                                                                                                                                                                                                                                                                    |
| <b>Scope</b>                                                                                | This unit covers the following; <ul style="list-style-type: none"> <li>Identifying Training Needs.</li> <li>Developing Training Materials.</li> <li>Coordinating All Training Programmes.</li> </ul>                                                                                                                                                                                                                                          |
| <b>Performance Criteria (PC) with respect to the Scope</b>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Element</b>                                                                              | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Identifying Training Needs</b>                                                           | To be competent, the individual must be able to:<br>PC1. Conduct training-needs assessments and identify staff skills gaps in food quality and safety aspects.                                                                                                                                                                                                                                                                                |
| <b>Developing Training Materials</b>                                                        | To be competent, the individual must be able to:<br>PC2. Design training materials on food quality and safety aspects.<br>PC3. Develop training programmes.                                                                                                                                                                                                                                                                                   |
| <b>Coordinating All Training Programmes</b>                                                 | To be competent, the individual must be able to:<br>PC4. Train and coordinate all food safety and quality training.<br>PC5. Coordinate with relevant stakeholders on staff related training programmes.                                                                                                                                                                                                                                       |
| <b>Knowledge and Understanding (K)</b>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>A. Organisational Context (Knowledge of the company/ organisation and its processes)</b> | The individual on the job must demonstrate knowledge and understanding of the organisational:<br>OK1. Food safety policies, rules, and regulations.<br>OK2. Standard operating procedures.<br>OK3. Food product requirements and specifications.<br>OK4. Equipment calibration and maintenance policy.<br>OK5. Hierarchy or organogram.                                                                                                       |
| <b>B. Technical Knowledge</b>                                                               | The individual on the job must demonstrate knowledge and understanding of:<br>TK1. Training-needs assessment skills.<br>TK2. Food safety and quality protocols, rules, and regulations.<br>TK3. Food production and processes skills.<br>TK4. Food deficiencies.<br>TK5. Food microbiology.<br>TK6. Hygiene and sanitation skills.<br>TK7. Food packaging skills.<br>TK8. Training skills.<br>TK9. Entrepreneurship.<br>TK10. Food chemistry. |



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| <b>C. Regulatory Context (knowledge of rules and regulations)</b> | <p>The individual on the job must demonstrate knowledge and understanding of the:</p> <p>RK1. Food and Drugs regulations (SI 90 of 2001).<br/>           RK2. Food Safety Act and other regulations.<br/>           RK3. Zambia Standards Act and other regulations.<br/>           RK4. Zambia Compulsory Standards Agency Act.<br/>           RK5. Local Government Act and other regulations.<br/>           RK6. Codex Alimentarius.<br/>           RK7. Occupational Health and Safety Act and other regulations<br/>           RK8. Factories Act.</p> |
| <b>Skills (S)</b>                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>A. Core Skills/ Generic Skills</b>                             | <b>Writing Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                   | <p>The individual on the job must be able to:</p> <p>CS1. Write in English language.<br/>           CS2. Prepare operators' instructions for subordinates.<br/>           CS3. Construct a non-compliance report.</p>                                                                                                                                                                                                                                                                                                                                        |
|                                                                   | <b>Reading Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                                   | <p>The individual on the job must be able to:</p> <p>CS4. Read English language.<br/>           CS5. Read and interpret results or instructions provided for the required work.</p>                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                   | <b>Oral Communication (Listening and Speaking Skills)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                   | <p>The individual on the job must be able to:</p> <p>CS6. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers.<br/>           CS7. Convey information clearly and concisely to co-workers.</p>                                                                                                                                                                                                                                                                                                             |
| <b>B. Professional Skills</b>                                     | <b>Decision-Making</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                   | <p>The individual on the job must be able to have:</p> <p>PS1. Training skills.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                   | <b>Plan and Organise</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                   | <p>The individual on the job should be able to plan and organise:</p> <p>PS2. Training schedules.<br/>           PS3. Procurement of training materials.</p>                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                   | <b>Customer Centricity</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                   | <p>The individual on the job should be able to:</p> <p>PS4. Coordinate with other internal units that relate directly with customer care with intent to satisfying their product requirements.</p>                                                                                                                                                                                                                                                                                                                                                           |
|                                                                   | <b>Problem-Solving</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                   | <p>The individual on the job should be able to:</p> <p>PS5. Resolve any conflicts within the team.<br/>           PS6. Upgrade training materials to latest available techniques and regulations.<br/>           PS7. Resolve any conflicts with clients in relation to training.</p>                                                                                                                                                                                                                                                                        |

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|  | <b>Analytical Thinking</b>                                                                    |
|  | The individual on the job should be able to:                                                  |
|  | PS8. Analyse and solve training deficiencies.                                                 |
|  | PS9. Convey to the superior any non-conformities and possible solutions.                      |
|  | PS10. Discuss with superior any occurrence that may require changing the training programmes. |
|  | <b>Critical Thinking</b>                                                                      |
|  | The individual on the job should be able to:                                                  |
|  | PS11. Identify and deal with or report deviations in training programmes to superiors.        |

**UNIT 6** [This Unit covers the skills and knowledge required by a Food Scientist to explore the field of entrepreneurship]

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| <b>Unit No.</b>                                            | <b>06</b>                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Unit Title</b>                                          | <b>Entrepreneurship for a Food Scientist</b>                                                                                                                                                                                                                                                                                                                                                |
| <b>Description</b>                                         | This Unit describes the knowledge and skills required by a Food Scientist to explore the field of entrepreneurship.                                                                                                                                                                                                                                                                         |
| <b>Scope</b>                                               | <p>This unit covers the following;</p> <ul style="list-style-type: none"> <li>• Consultation in Setting Up of Food Laboratories.</li> <li>• Setting Up Training Centres in Food Science.</li> <li>• Establishing Private Food Testing Laboratories.</li> <li>• Marketing Laboratory Equipment and Materials.</li> <li>• Setting Up a Food or Beverage Product Plant.</li> </ul>             |
| <b>Performance Criteria (PC) with respect to the Scope</b> |                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Element</b>                                             | <b>Performance Criteria (PC)</b>                                                                                                                                                                                                                                                                                                                                                            |
| <b>Consultation in Setting Up of Food Laboratories</b>     | <p>To be competent, the individual must be able to:</p> <p>PC1. Identify opportunities where specific expertise can be provided in the field of food science.</p> <p>PC2. Determine relevant areas of economic importance in the field of food production.</p>                                                                                                                              |
| <b>Setting Up Training Centres in Food Science</b>         | <p>To be competent, the individual must be able to:</p> <p>PC3. Determine areas of significant relevance to the food industry projecting into future years.</p> <p>PC4. Develop training materials for students at various tertiary levels.</p> <p>PC5. Operate analytical equipment related to the training.</p>                                                                           |
| <b>Establishing Private Food Testing Laboratories</b>      | <p>To be competent, the individual must be able to:</p> <p>PC6. Determine areas of food science relevant to the immediate community and abroad.</p> <p>PC7. Manage subordinate Food Scientists and support staff.</p> <p>PC8. Acquire and maintain business acumen.</p> <p>PC9. Acquire ISO 17025 or relevant accreditation of the laboratory.</p>                                          |
| <b>Marketing Laboratory Equipment and Materials</b>        | <p>To be competent, the individual must be able to;</p> <p>PC10. Determine the current trends in food production and testing.</p> <p>PC11. Identify the relevant equipment in the industry.</p> <p>PC12. Acquire marketing skills.</p>                                                                                                                                                      |
| <b>Setting Up a Food or Beverage Product Plant</b>         | <p>To be competent, the individual must be able to;</p> <p>PC13. Determine business viability in chosen location.</p> <p>PC14. Identify the critical ingredients of the specific products.</p> <p>PC15. Enhance product nutritional content within confines of quality specifications and regulations.</p> <p>PC16. Coordinate with Marketing department to meet customer satisfaction.</p> |

| <b>Knowledge and Understanding (K)</b>                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>A. Organisational Context<br/>(Knowledge of the company/ organisation and its processes)</b> | <p>The individual on the job must demonstrate knowledge and understanding of:</p> <p>OK1. Organisational structures of food safety company/entity, policies, rules, and regulations (locally and internationally)</p> <p>OK2. Relevant national standards, policies, and procedures followed in the registration of the business/company.</p> <p>OK3. Business/company roles, responsibilities, accountabilities and authorities.</p>                                                                                              |
| <b>B. Technical Knowledge</b>                                                                   | <p>The individual on the job must demonstrate knowledge and understanding of:</p> <p>TK1. Creating a business plan for a food safety company.</p> <p>TK2. Acquiring resources and financing for a new business.</p> <p>TK3. Hiring suitable labour for a food safety company.</p> <p>TK4. Providing leadership and management for a food safety company.</p>                                                                                                                                                                       |
| <b>C. Regulatory Context<br/>(Knowledge of rules and regulations)</b>                           | <p>The individual on the job must demonstrate knowledge and understanding of the:</p> <p>RK1. HACCP</p> <p>RK2. GMP's</p> <p>RK3. FSSC 2200</p> <p>RK4. Food and Drugs regulations (e.g. SI 90 of 2001).</p> <p>RK5. Food Safety Act and other regulations</p> <p>RK6. Zambia Standards Act and other regulations.</p> <p>RK7. Zambia Compulsory Standards Agency Act.</p> <p>RK8. Local Government Act and other regulations.</p> <p>RK9. Codex Alimentarius.</p> <p>RK10. Patents and Companies Registration Agency (PACRA).</p> |
| <b>Skills (S)</b>                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>A. Core Skills/<br/>Generic Skills</b>                                                       | <b>Writing Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                 | <p>The individual on the job must be able to:</p> <p>CS1. Communicate in writing with various client types.</p> <p>CS2. Construct and interpret non-compliance reports in laboratory setups or consultations.</p>                                                                                                                                                                                                                                                                                                                  |
|                                                                                                 | <b>Reading Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                 | <p>The individual on the job must be able to:</p> <p>CS3. Read and understand relevant documentation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                                                 | <b>Oral Communication (Listening and Speaking Skills)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                                                 | <p>The individual on the job must be able to:</p> <p>CS4. Listen attentively and interpret communication from various clientele and subordinates.</p>                                                                                                                                                                                                                                                                                                                                                                              |
| <b>B. Professional Skills</b>                                                                   | <b>Decision-Making</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                                 | <p>The individual on the job must be able to:</p> <p>PS1. Decide on relevant business undertakings set in specific strategic plans.</p>                                                                                                                                                                                                                                                                                                                                                                                            |

|  |                                                                                                                                                    |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Plan and Organise</b>                                                                                                                           |
|  | The individual on the job should be able to:                                                                                                       |
|  | PS2. Plan and organise a business entity.                                                                                                          |
|  | PS3. Coordinate marketing strategies related to the business entity.                                                                               |
|  | <b>Customer Centricity</b>                                                                                                                         |
|  | The individual on the job should be able to:                                                                                                       |
|  | PS4. Coordinate with key stakeholders that relate directly with the business entity with intent to satisfying their product/ service requirements. |
|  | PS5. Apply business code of ethics and good manufacturing practices with emphasis on consumer protection.                                          |
|  | <b>Problem-Solving</b>                                                                                                                             |
|  | The individual on the job should be able to:                                                                                                       |
|  | PS6. Resolve any conflicts with clients to provide excellent product/service delivery.                                                             |
|  | <b>Analytical Thinking</b>                                                                                                                         |
|  | The individual on the job should be able to:                                                                                                       |
|  | PS7. Analyse and apply constructive business strategies.                                                                                           |
|  | <b>Critical Thinking</b>                                                                                                                           |
|  | The individual on the job should be able to:                                                                                                       |
|  | PS8. Identify and deal with areas that tend to derail strategies laid down for business profit.                                                    |

## 5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to:

**Equipment and Tools:** Design and prototyping equipment and tools; computer software applications; food production equipment; packaging equipment; maintenance equipment and tools; testing equipment and tools; personal protective equipment, etc.

**Raw Materials and Consumables:** Food ingredients, etc.

## 6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES

Food Scientists face challenges such as obsolete and/or inappropriate equipment and tools; budgetary constraints; inadequate product costing skills; poor technical skill base; bureaucracy in procurement procedures; lack of appreciation of preventive maintenance by non-engineering management staff; labour intensive nature of the work; rapid change of technology and materials; lack of personal protective equipment; climate change; cyber warfare; inconsistency in company and government policies and regulations, etc.

### 6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to challenges would include: selecting and procuring appropriate equipment and tools for the job; supporting capacity building through training; identifying and utilising suitable adaptation and mitigation measure against the effects of climate change; utilising appropriate cyber security measures to protect against cyber warfare; inclusion of engineering professionals in management teams; deployment of automation where feasible; provision of personal protective equipment; participating in lobbying and formulation of policies; allocation of adequate financial resources, etc.

## 7. WORKING CONDITIONS/ENVIRONMENT

Food Scientists work with a variety of machinery, toxic substances, and volatile materials. Their work environment is susceptible to fires, explosions, structural failures, and equipment malfunctions. Working conditions include: cold, hot, and wet conditions; climbing great heights; standing/walking for long hours; lifting materials; working in day or night shifts; areas that are noisy and dusty; areas with limited lighting or ventilation, etc.

## 8. PARTIES INVOLVED/INTERACTING WITH THE JOBHOLDER OR TRAINEE

### 8.1 Internal/Within the Organisation

Management, supervisors, subordinates, and other section members, etc.

## **8.2 External/Outside the Organisation**

Government regulators, professional bodies, clients, suppliers, fellow Food Scientists from other companies, labour unions, students/interns, etc.

## **9. PHYSICAL DEMANDS ON THE BODY**

- Physique to sustain strenuous conditions;
- Be able to walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Be able to use fingers, hands, and feet with ease to complete the assigned task (dexterity).

## **ANNEX A**

### **Criteria for Assessments based on this NOS**

#### **A.1 Guidelines for Assessment**

**A.1.1** Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programme developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programme developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'Out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

**A.1.2** Individual awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part, and evaluations for skill practical part for their respective candidates.



## ANNEX B

### NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

|                   |                 |                            |                |
|-------------------|-----------------|----------------------------|----------------|
| <b>NOS Code</b>   | NOS.FS.01       |                            |                |
| <b>ZQF Level</b>  | 7               | <b>Version Number</b>      | 01             |
| <b>Sector</b>     | Manufacturing   | <b>Date of Approval</b>    | February, 2024 |
| <b>Sub Sector</b> | Food Processing | <b>Date of Last Review</b> | N/A            |
| <b>Occupation</b> | Food Science    | <b>Date of Next Review</b> | February, 2029 |



*“You learn,  
We Standardise”*

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